Fiscal Year:	FY 2017	**Task Last Updated:**	FY 10/25/2017
PI Name:	Haswell, Elizabeth Ph.D.		
Project Title:	Mechanosensitive Ion Channels in Plants: Genetic, Computational and Systems-Level Approaches to Understanding their Proposed Role in Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	63130-4862	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	October 2017 final report: Matt Geer--paid summer intern on fellowship ; Matt Mixdorf, Technician -- 50% from 4/1/16 to 2/28/17 ; 100% from 3/1/17 to 6/9/17. Debarati Basu, Postdoc -- 100% from 5/1/17 to 7/1/17; 50% from 7/1/17 to 8/1/17. Eric Schultz, Postdoc -- 50% from 8/1/16 to 1/31/17; 100% from 2/1/17 to 4/28/17.		
COI Name (Institution):	Spalding, Edgar Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX13AM55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	How plants respond to gravity is a fundamental biological question with important implications for spaceflight. Here we propose ground-based research designed to lead to spaceflight, in the area of plant biology. Our experimental approach focuses on several families of putative mechanoreceptors known as mechanosensitive (MS) ion channels. MS ion channels have been proposed to play a role in the perception of gravity in plants for many years, a hypothesis that is supported by several lines of evidence, but one that has not yet been systematically tested by molecular genetics. The research plan utilizes contemporary methodologies for the study of the model plant Arabidopsis thaliana to interrogate the biological function of plant MS ion channels and builds upon previous NASA-funded research into the response of plants to altered gravity environments. We will determine the role played by MS ion channels in mediating gravity signal transduction, their putative roles in adaptation to microgravity, and the molecular mechanisms by which the model MS ion channel MSL10 activates stress-response pathways in plant cells. We expect that these activities will have the following outcomes: 1) Test the long-standing hypothesis that MS ion channels function in gravity perception; 2) Contribute to our basic knowledge regarding how plants exploit MS ion channels to sense and respond to mechanical stress; 3) Provide information that could be used on Earth to improve crop productivity; and 4) Establish a foundation of knowledge required to generate more mechanistic and sophisticated hypotheses about plant gravity perception that can be tested in spaceflight using the European Modular Cultivation System and/or the Advanced Biological Research System (ABRS) Green Fluorescent Protein (GFP) Imaging system. (These proposed studies directly address the objectives of the solicitation, as they aim to answer basic questions about how plants respond to changes in gravity and should lead to hypotheses that can be tested on future spaceflights.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identifying the molecular mechanisms by which MS ion channels function and evaluating their roles in the perception and response to altered gravity conditions will help fill a large gap in our knowledge regarding the molecules involved in gravity and mechanoperception. Gaining this knowledge has the potential to achieve two long-term goals: 1) to mitigate the effects of mechanical stress on crop plants on Earth, and 2) to counteract the effects of altered gravity environments on plants grown in space. These experiments will address the high priority goal of “understanding the basic mechanisms of plant response” to changes in gravity described in the Decadal survey of 2011.
Task Progress:	Research Progress May 2016-July 2017 Aim 1: Test the Hypothesis that MS ion channels Play a Role in Root Gravitropism Using a Machine Vision and Computation Approach. Mechanosensitive (MS) ion channels have been implicated in a variety of steps in gravity signaling, but are most often implicated in the initial gravity response. MS ion channels have been proposed to sense ER membrane tension as a result of the amyloplasts sedimentation; the plasma membrane tension resulting from the uneven pressure of the tonoplast at the bottom of the cell; or the action of amyloplasts on actin microfilaments tethered to the plasma membrane. There are at least three potential barriers to the identification of MS ion channels in forward genetic screens for genes involved in gravity response: 1) redundancy of MS ion channel gene function; 2) redundancy of gravity response pathways involving MS ion channels; and 3) subtle effects of MS ion channel function on the gravity response. We aim to test the long-held hypothesis that MS ion channels mediate the perception of changes in gravity vector using a reverse genetics approach and a highly sensitive Machine Vision and Computation assay for defects in gravity perception. Progress: We further characterized the gravity defect in the msl2 msl3 mutants. The msl2 msl3 double mutant lacks MS channels in the plastid envelope. While the msl2 msl3 mutant shows overall defects in orientation when grown vertically on solid media, or on a re-positioning machine, there are developmental defects that complication interpretation. Several aspects of cellular morphology that are key for gravity response disappear or appear disorganized in msl2 msl3 mutants compared to those in wild type seedlings. We tested a range of media with added osmotic support and saw that we could suppress these phenotypes to some degree, but in the end were unable to generate growth conditions where root growth was normal enough to feel confident that defects in gravity could be attributed to perception rather than to anatomical differences. We worked towards generating msl9 msl10 pgm and msl4 msl5 msl6 pgm mutants. We tested several AtPiezo CRISPR/Cas9 gene knockouts for gravity defects but did not observe any. Aim 2: Test the Hypothesis that MSL and other MS ion channels are involved in the perception and/or Adaptation to Weightlessness. Progress: We do not have progress to report on this aim. Aim 3: Determine the Molecular Mechanism of MSL10-Induced Reactive Oxygen Species (ROS) Signaling. MS ion channels are implicated in the perception and/or response to changes in gravity conditions. The experiments in this aim are designed to gain a more molecular and mechanistic handle on the regulation and signaling behavior of MS ion channels, building on our recent observations that overexpressing the MscS-Like MS ion channel MSL10 from A. thaliana leads to ROS-associated cell death. Understanding how MSL10 is regulated and how its activation leads to ROS-related stress signaling may be important for mitigating the effects of mechanical stress on crop plants or plants in the space environment. Progress: A. We established that endogenous expression of unphosphorylated MSL10 and a single point mutation the MSL10 C-terminus result in the hyper-accumulation of ROS. Furthermore, the chaperone complex SGT1/RAR1/HSP90 is required for this phenomenon. Previous years we showed that overexpression of MSL10 leads to cell death signaling and the production of ROS at endogenous levels of expression. Now we can say that dephosphorylation-dependent cell death signaling does not require overexpression of MSL10, as we observe cell death, dwarfing, and the production of ROS in plants expressing phospho-dead MSL10, but not phospho-mimic MSL10 at endogenous levels. Furthermore, a single EMS-induced point mutation in the MSL10 C terminus, S640L, produces the same phenotypes. HSP90 and its co-chaperone SGT1 play a conserved role in plant and animal innate immunity by stabilizing pathogen sensors called NLR proteins. It was also recently reported that HSP90 and SGT1 stabilize a hormone receptor in response to heat stress, indicating that this chaperone complex plays a role in both biotic and abiotic stress responses.

	We found that HSP90, SGT1, and RAR1 are all required for MSL10-induced cell death in the transient tobacco expression assay, and for the hyper-accumulation of ROS, indicating that the same is true in stably transformed plants. However, we do not detect direct interactions between MSL10 and any of these components by a variety of methods. B. We determined that 14-3-3 proteins interact with MSL10 in a specific manner. 14-3-3 proteins are evolutionarily conserved proteins that often recognize and bind to the loose consensus sequences R/KXXS or R/KXXXS, subsequently regulating the function of their binding partners or “clients” via effects on enzyme activity, subcellular localization, protein-protein interaction, conformation, or stability. Four of the seven in vivo phosphorylated residues in the MSL10 N-terminus are part of a 14-3-3 consensus binding site. While full-length MSL10 interacts with all 14-3-3 proteins in the split-ubiquitin yeast two hybrid, the introduction of four point mutations designed to disrupt the 14-3-3 consensus sequences at S48, S57, S128, and S131 also disrupt the interaction. As is shown, MSL10K45A K53A K124A R127A interacts normally with the positive control in this assay. Manuscripts in Preparation: D. Basu, K. M. Veley, J. Shoots, Matt Mixdorf and E. S. Haswell. MSL10 induces cell death through SGT1/RAR1/HSP90 and 14-3-3 proteins in response to mechanical disturbance at plasma membrane. Submission target: Spring 2018.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Wilson ME, Mixdorf M, Berg RH, Haswell ES. "Plastid osmotic stress influences cell differentiation at the plant shoot apex." Development. 2016 Sep 15;143(18):3382-93. https://doi.org/10.1242/dev.136234 ; PubMed PMID: 27510974; PubMed Central PMCID: PMC5047659 , Sep-2016
Articles in Peer-reviewed Journals	Basu D, Haswell ES. "The mechanosensitive ion channel MSL10 potentiates responses to cell swelling in Arabidopsis seedlings." Curr Biol. 2020 Jul 20;30(14):2716-28.e6. https://doi.org/10.1016/j.cub.2020.05.015 ; PMID: 32531281. , Jul-2020